WEEK 1 · 9–12 NOVEMBER 2026

Preprocessing concepts for hyperspectral data

Which pixels and which bands deserve your trust

Week 1 · Session 5 · practical: 11_preprocessing.ipynb

What you will be able to do

- Place a product on the chain radiance → TOA reflectance → surface reflectance
- Explain what atmospheric correction removes, and what it leaves behind
- Inspect quality masks critically, and build your own test when they fail
- Justify which bands you drop, using the flag, the SNR and residual features
- Apply smoothing, continuum removal and brightness normalisation without damaging the signal
- Cut, subset and export data so your project runs fast and travels well

1. Processing levels

From photons at the sensor to reflectance of the ground

What the number in the file means

Level	Quantity	What has been done	Tanager asset
Radiance	L(λ), W m^{-2} sr^{-1} μm^{-1}	calibrated counts, as seen at the satellite	ortho_radiance_hdf5
TOA reflectance	ρ_TOA, 0–1	radiance ÷ incoming sunlight	computed by the user
Surface reflectance	ρ_s, 0–1	atmospheric absorption and scattering removed	ortho_sr_hdf5

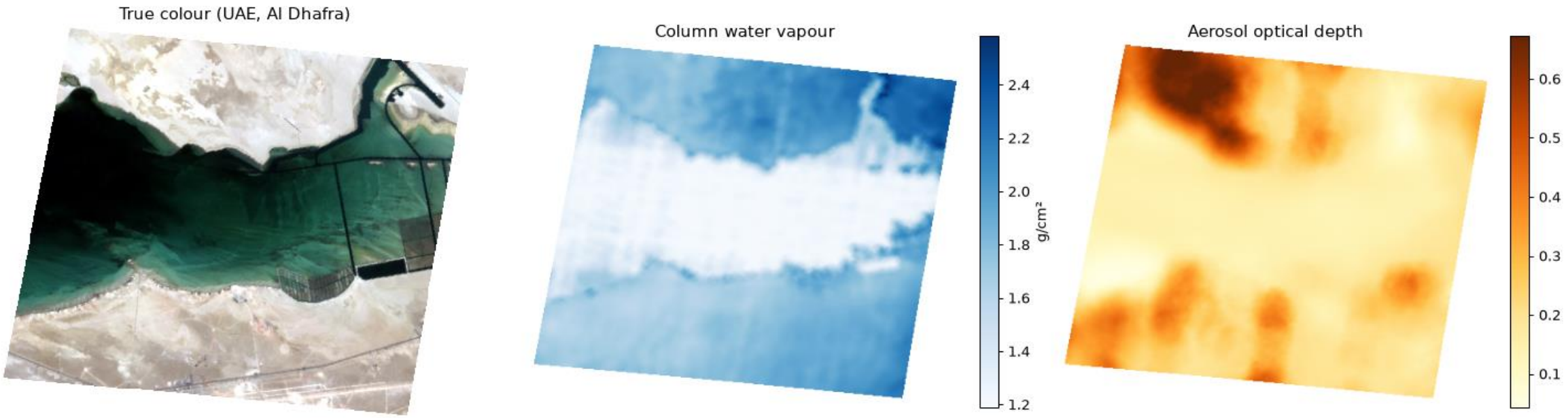
Basic or ortho?

- **Basic** products stay in the sensor's viewing geometry; a separate geolocation array gives each pixel's lon/lat
- **Ortho** products are terrain-corrected and resampled onto a map grid — UTM, 30 m for Tanager
- Orthorectification uses a DEM and the sensor model to remove relief and viewing distortions
- Use ortho unless you have a reason not to, such as avoiding resampling before a physics retrieval
  - The rotated swath inside a north-up grid is why 20–30 % of every scene is no-data

What the atmosphere does to your signal

- **Absorbs**: water vapour (940, 1130, 1380, 1880 nm), O_2 (760 nm), CO_2 (2010, 2060 nm), CH_4 (2300 nm)
- **Scatters**: molecules (Rayleigh, strongest in the blue) and aerosols — haze, dust, smoke
- Scattering adds *path radiance*: light that never touched the ground but reaches the sensor
- Radiative-transfer codes (MODTRAN, 6S, the open-source ISOFIT used for EMIT) model both effects
- They need per-pixel **column water vapour** and **aerosol optical depth** — Tanager ships both as layers

The atmosphere varies inside a single scene



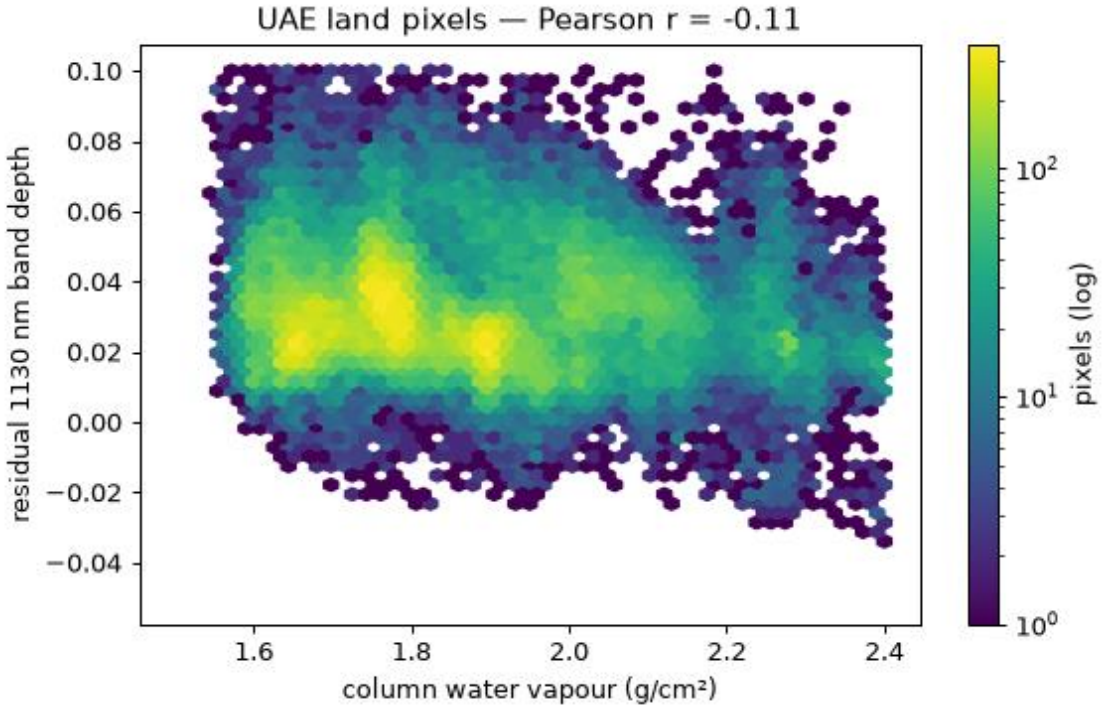
Al Dhafra, UAE — true colour, column water vapour (1.21–2.16 g/cm²), aerosol optical depth (0.12–0.54)

Read the ancillary layers critically

- The water-vapour map jumps at the **coastline** — an artefact of the retrieval over dark water, not weather
- The highest AOD sits over the **brightest sabkha**, where separating aerosol from surface is hardest
- These layers are retrievals with their own errors, not ground truth
- Use them as **quality covariates**: is my anomaly correlated with AOD, or does it follow the coastline?

Did the correction work?

- Test: the residual depth of the 1130 nm water-vapour feature should not track column water vapour
- Over UAE land the correlation is essentially flat
- Median residual depth is **0.030** — small, but systematic
- You will see small dips near 940 and 1130 nm in almost every spectrum
- **Do not** interpret them as surface features



Residual 1130 nm band depth vs column water vapour, UAE land pixels

2. Quality masks

Inspect before you trust

THE LESSON OF THIS SESSION

A mask is somebody else's classification of your data.

Tanager's cloud and cirrus masks are labelled *beta*. On the Al Dhafra scene, a cloud-free day, the cloud mask flags 40 % of the grid — and 97 % of the bare land.

WHAT YOU GET

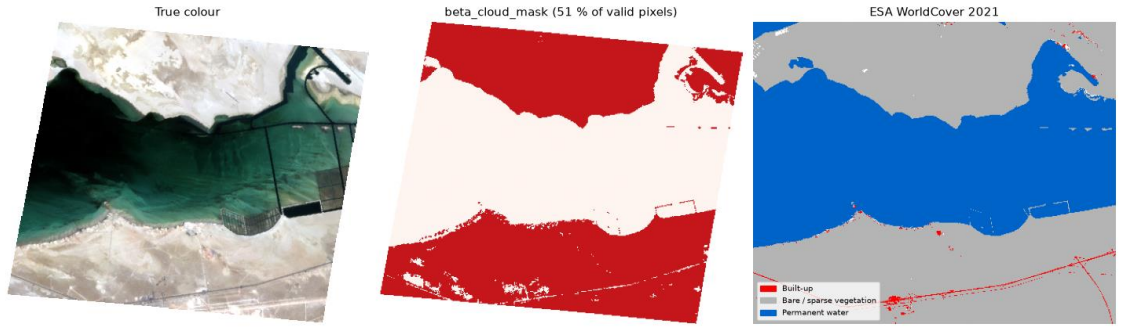
The three quality layers in every Tanager file

Layer	Meaning
nodata_pixels	outside the imaged footprint — the rotated strip leaves 20–30 % of the grid empty
beta_cloud_mask	cloud, from Planet's beta cloud classifier
beta_cirrus_mask	thin high cirrus

AL DHAFRA, UAE

False positives: bright ground read as cloud

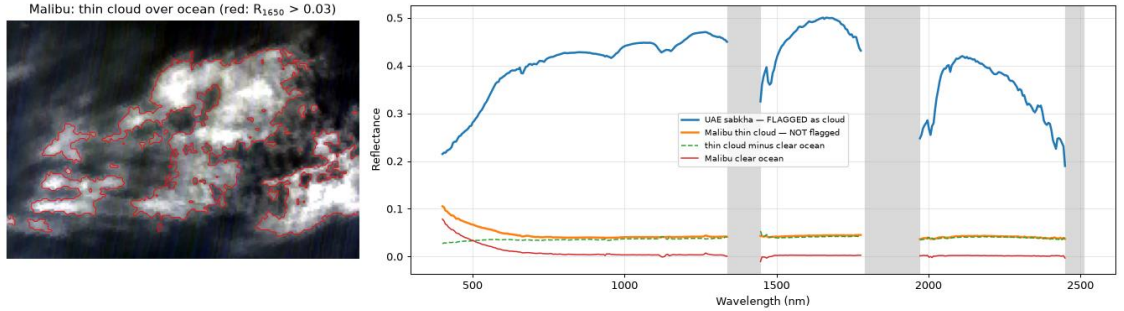
- Al Dhafra, 11 May 2025 — a clear scene
- **97.1 %** of bare/sparse vegetation flagged as cloud
- **63.2 %** of built-up, but only **2.2 %** of water
- Bright white surfaces are the classic failure mode of cloud detectors
- Blind use would delete half the land in the scene



True colour · beta cloud mask · ESA WorldCover classes

False negatives: thin cloud the mask misses

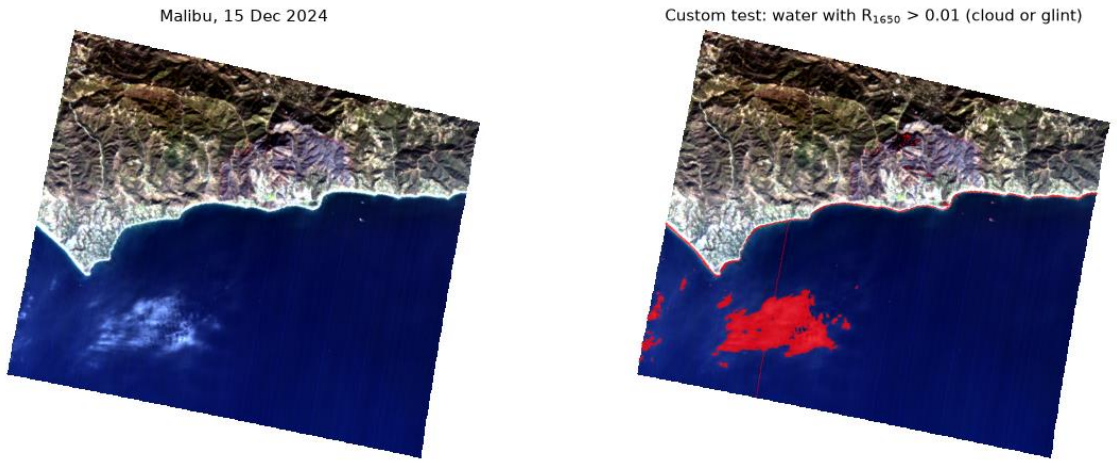
- Malibu, 15 Dec 2024 — only **36 pixels** flagged in the entire scene
- Thin cloud over the ocean adds a nearly flat offset, including in the SWIR
- Clear water is almost black beyond 1000 nm
- Flagged sabkha, by contrast, shows real mineral structure
- **Spectra let you audit a mask** pixel by pixel



Thin cloud over the ocean vs clear ocean vs flagged sabkha

Build the test your target needs

- Rule: water pixels with R(1650 nm) > 0.01 are cloud or glint
- Finds **16,682** suspect water pixels where the beta mask found 36
- It also catches the **surf zone** — add a coastal buffer
- And a faint vertical **detector stripe** from the pushbroom array
- Artefacts are only visible over uniform, dark surfaces



Custom SWIR cloud/glint test on the Malibu scene

Your masking decision, written down

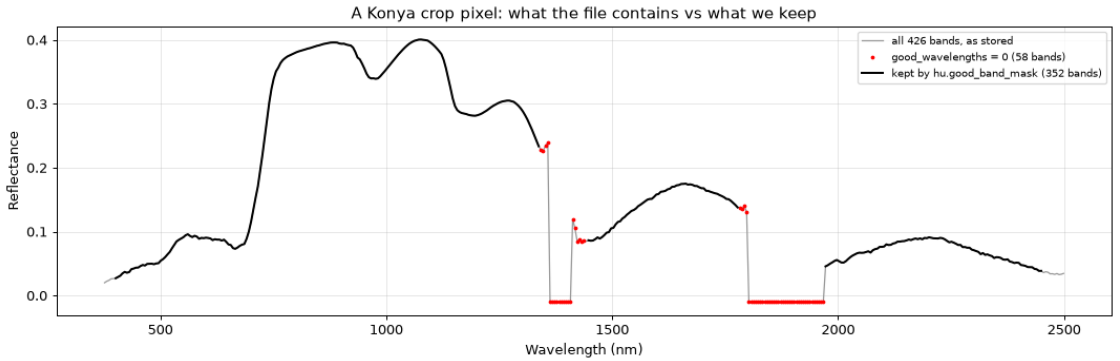
- Use the masks as delivered (`cloud=True`) when they are right for your scene
- Ignore them (`cloud=False`) when they fail — as in the UAE sabkha scene
- Or build a target-specific rule, like the SWIR water test, and document it
- **Always** overlay the mask on an RGB image before deciding
- Record the decision and the reason in your project report — a reviewer will ask

3. Bad bands

Which of the 426 bands carry information?

The flag and the placeholders

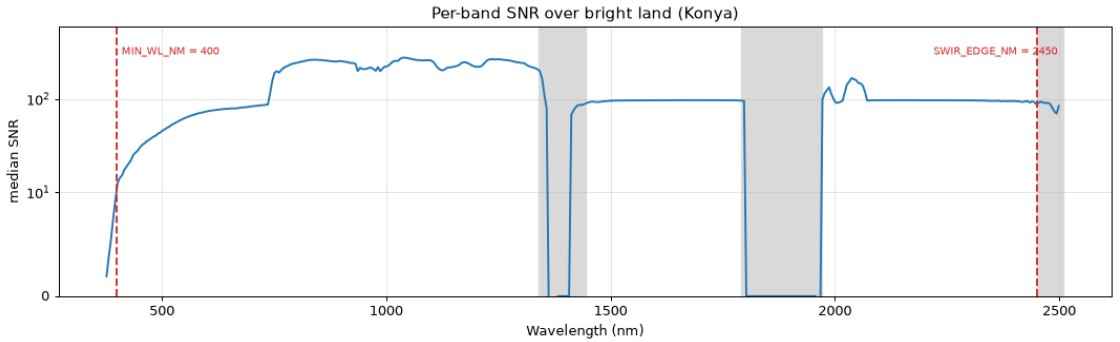
- The file carries a per-band `good_wavelengths` flag: 0 = unusable
- 58 bands are flagged, in the deep water-vapour windows
- In the core of those ranges the file stores a constant **–0.01**, not a measurement
- 44 of the 58 flagged bands hold that placeholder
- Edge bands look plausible but are unreliable



A Konya crop pixel: all 426 bands, the flagged bands, and the 352 bands we keep

Signal-to-noise decides the ends of the spectrum

- SNR per band = reflectance ÷ the file's uncertainty layer
- Below 400 nm the SNR runs **1.9, 3.8, 5.4, 7.5, 9.3** — unusable
- Little UV reaches the ground and Rayleigh scattering is strongest there
- Beyond 2450 nm the window closes and edge detectors are least reliable
- SNR also collapses at the edges of the water-vapour windows

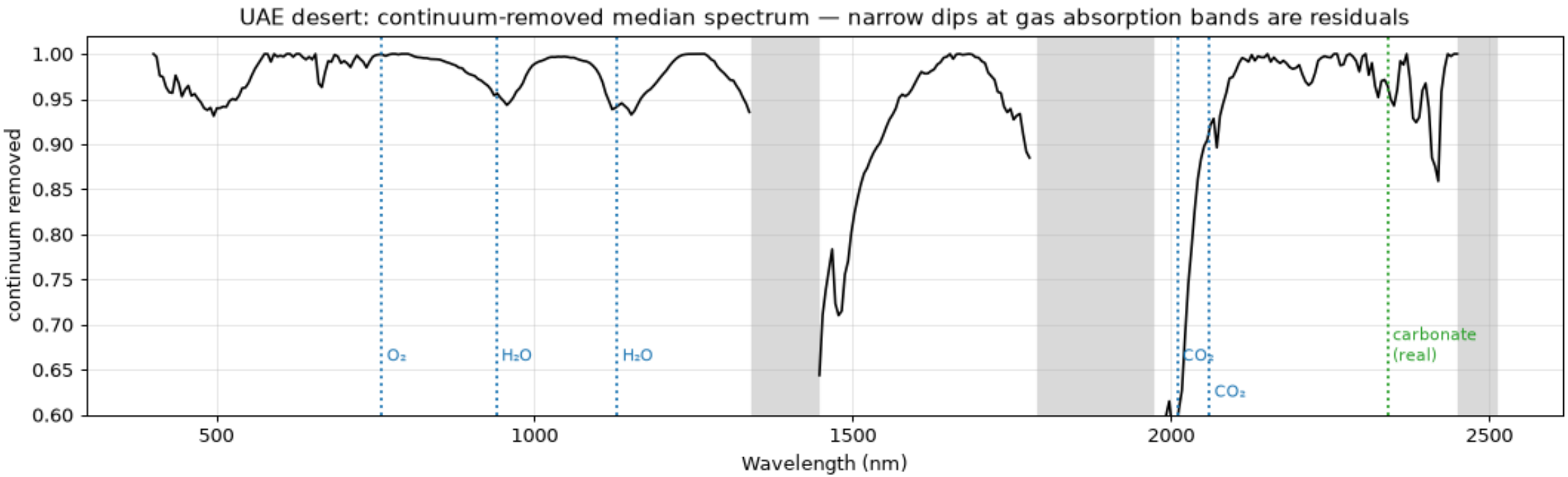


Median SNR over bright land, Konya

The three cuts applied by `bands="good"`

- The two **water-vapour windows**, slightly wider than the file flag: 1340–1445 and 1790–1970 nm
- Everything **below 400 nm** — SNR under 10
- Everything **above 2450 nm** — erratic spectra at the detector edge
- 426 bands in, **352 bands** out
  - Add your own ranges with `extra_bad=((1440, 1490), (1995, 2020))` for water studies

What the correction left behind



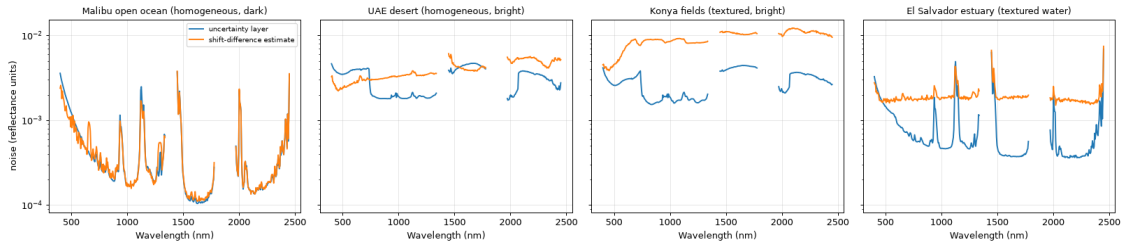
UAE desert, continuum-removed median spectrum — narrow dips at gas absorption bands are residuals

4. Noise, smoothing and features

Improving the signal without inventing it

Two independent noise estimates

- **Uncertainty layer**: the provider's per-band estimate
- **Shift difference**: $\sigma \approx 1.4826 \cdot \mathrm{MAD}(x_{i+1} - x_i) / \sqrt{2}$ over a homogeneous patch
- Over open ocean the two agree — noise ≈ 0.0002 reflectance
- Over desert they agree within about a factor of two
- Over fields and sediment plumes the shift estimate is several times larger



Four targets, two estimators

Savitzky-Golay smoothing, done correctly

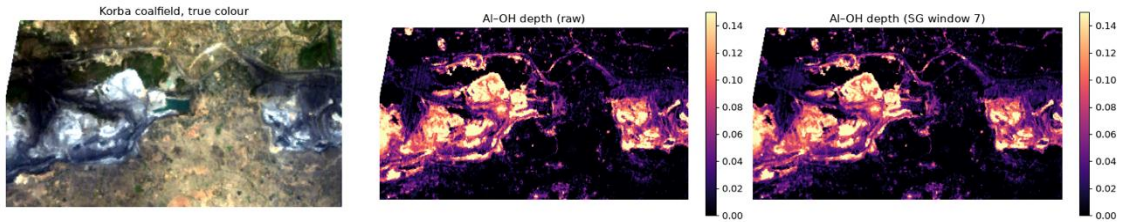
- Fits a low-order polynomial in a moving window along the spectrum (Savitzky & Golay, 1964)
- Keeps feature position and depth better than a moving average
- **Trap 1:** never smooth across a removed gap — 1335 and 1450 nm are not neighbours
- **Trap 2:** `savgol_filter` fails on NaN, so masked pixels need temporary filling
- With ~5 nm sampling, windows of 5–11 bands (25–55 nm) are sensible
- Choose the **smallest** window that gives stable results

When not to smooth

- When **narrow features are the signal**: mineral doublets, gas absorption for methane retrieval
- Before **noise-dependent methods**: MNF, matched filters, covariance estimation — smoothing changes the noise
- When comparing with **lab spectra** convolved to the sensor response
- Smoothing is a modelling choice: state the window and order in your report

From spectrum to map: band depth

- Continuum removal makes absorption depths comparable between bright and dark materials
- Applied per pixel, a band depth becomes a **mineral map**
- Al–OH depth at 2205 nm over the Korba coalfield
- Pit walls and clay-rich overburden light up; fields and forest stay dark
- Smoothing changes map noise only modestly here: 0.0140 → 0.0123



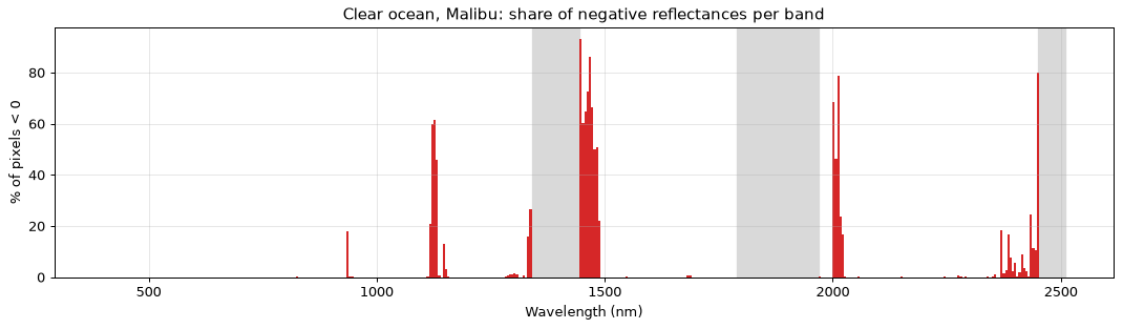
Korba coalfield — true colour, Al–OH depth raw, Al–OH depth smoothed

5. Water, geometry and space

Three practical decisions

Negative reflectance is not a bug to delete

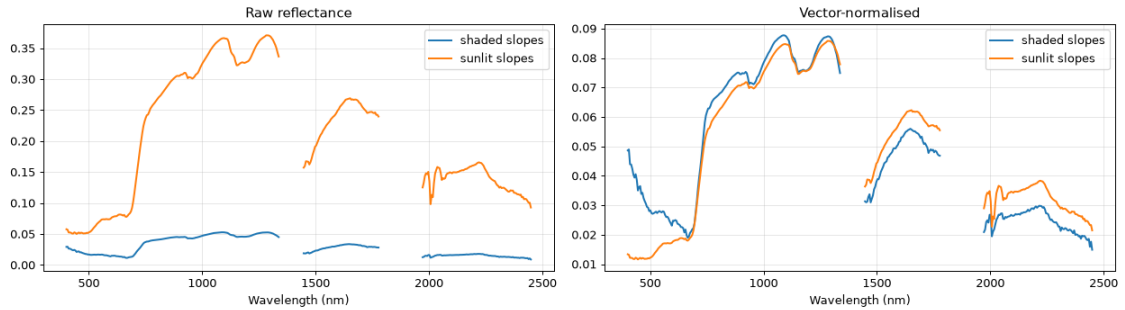
- Clear ocean, Malibu: **28 bands** have more than 10 % negative values
- Worst at 1448, 1468, 2449 and 2012 nm — absorption-window edges
- Setting negatives to NaN: complete water spectra fall from **100 % to 0 %**
- It also **biases** means upward: −0.0019 becomes +0.0016 at 2012 nm
- Keep them, or drop the offending bands; clip to zero only for display



Share of negative reflectances per band over clear ocean

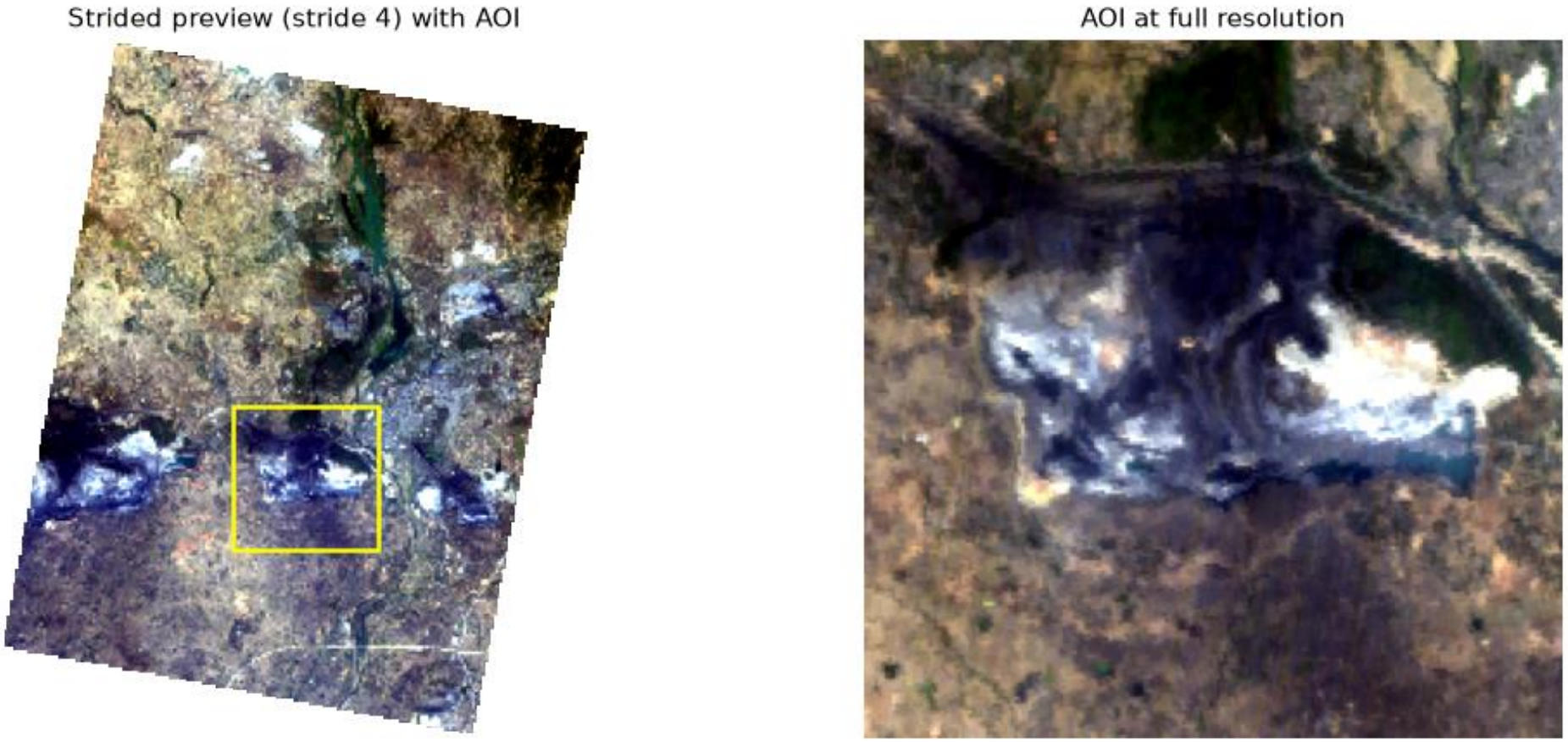
Illumination changes brightness, and a little of the shape

- Malibu in December: sun zenith ≈ 58.6°, strong topographic shading
- Sunlit vs shaded chaparral of similar greenness differ **6.7×** in NIR brightness
- Vector normalisation $\rho/\|\rho\|_2$ cuts Euclidean distance from 3.73 to 0.16
- The **spectral angle stays 9.3°** — SAM is invariant to brightness by construction
- Shaded slopes are lit by bluer diffuse skylight, so shape changes slightly too



Shaded and sunlit chaparral, raw and vector-normalised

Cut it down: preview, window, subset, export



Strided preview of the Korba scene with a 200 × 200 pixel AOI around the Kusmunda mine

Subset, export, reproject

- `hu.write_subset` stores a window as int16 × 10 000: **7.4 MB** against a **1.23 GB** scene
- Round-trip difference is 5×10^{-5} reflectance, and the georeferencing is identical
- Export results as a **Cloud-Optimised GeoTIFF** with band names — QGIS and web maps read it directly
- Reprojection **resamples**: analyse on the native grid, reproject only final products
  - Nearest neighbour for class maps, bilinear for continuous values

The whole chain, as one reusable function

```
def preprocess(scene, window=None, stride=1, use_cloud_mask=True,
               extra_bad=(), smooth_window=0, normalize=None):
    # use_cloud_mask : apply beta cloud/cirrus masks — inspect them first
    # extra_bad      : extra (start, stop) nm ranges to drop, e.g. for water
    # smooth_window  : Savitzky-Golay window in bands (0 = none)
    # normalize      : None, "vector" or "brightness"
    ...
    return cube, wavelengths, valid, report
```

The report dictionary records every choice, ready to paste into your method section.

PRACTICAL

Over to you

▶ `11_preprocessing.ipynb` · `120 min`

- Compare the cloud mask with WorldCover over the UAE scene and quantify the false positives
- Build the SWIR water test on the Malibu scene and find the detector stripe
- Plot per-band SNR and justify where you would cut the spectrum for your own theme
- Write a subset and export a COG, then open both again to check they are correct

KEY TAKEAWAYS

Recap

- Know your level: **surface reflectance** for analysis, radiance for gas retrievals
- Atmospheric correction leaves **residuals** near 760, 940, 1130 and 2010 nm
- Masks are **beta**: 97 % of UAE bare land flagged as cloud, thin cloud over Malibu missed
- Drop bands with evidence: the flag, the placeholders, SNR, residual features — 426 → 352
- Keep negative water reflectances; deleting them destroys water spectra and biases means
- Normalise brightness with care; smooth only when the noise, not the feature, dominates
- Subset early: 7.4 MB travels, 1.23 GB does not

Next: Week 2 — Hyperspectral image processing workflows: turning these decisions into a pipeline

References and further reading

Savitzky, A. & Golay, M.J.E. (1964). Smoothing and differentiation of data by simplified least squares procedures. Analytical Chemistry 36, 1627–1639.

Clark, R.N. & Roush, T.L. (1984). Reflectance spectroscopy: quantitative analysis techniques for remote sensing applications. JGR 89, 6329–6340.

Gao, B.-C., Montes, M.J., Davis, C.O. & Goetz, A.F.H. (2009). Atmospheric correction algorithms for hyperspectral remote sensing data of land and ocean. RSE 113, S17–S24.

Thompson, D.R. et al. (2018). Optimal estimation for imaging spectrometer atmospheric correction (ISOFIT). RSE 216, 355–373.

Green, A.A. et al. (1988). A transformation for ordering multispectral data in terms of image quality with implications for noise removal. IEEE TGRS 26, 65–74.

Course material: notebook 11_preprocessing.ipynb · quizzes/quiz_w1b_preprocessing.md